

Week 3

FedAvg and Baseline Federated Algorithms

From local SGD to round-based collaboration training

Graduate course · English instruction



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The Core Question

How do many clients train one model without sending their raw data to the server?

Local work

Each client updates a model using only its own local data.

Focused communication

Clients exchange model updates or gradients—not raw examples.

Aggregation

The server combines updates into the next global model.



FedSGD and FedAvg differ mainly in what happens during the local-training stage.

Learning Objectives

Training pipeline

Explain clients, server, local training, aggregation, and communication rounds.

FedSGD

Understand one-step gradient aggregation as a baseline.

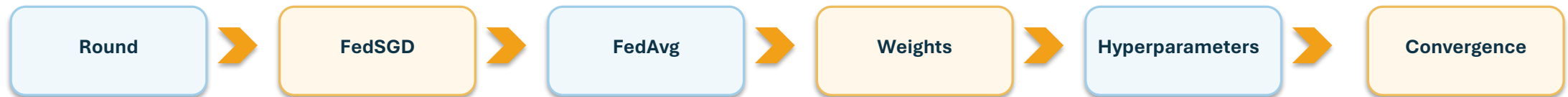
FedAvg

Derive weighted model averaging after multiple local steps.

Analysis

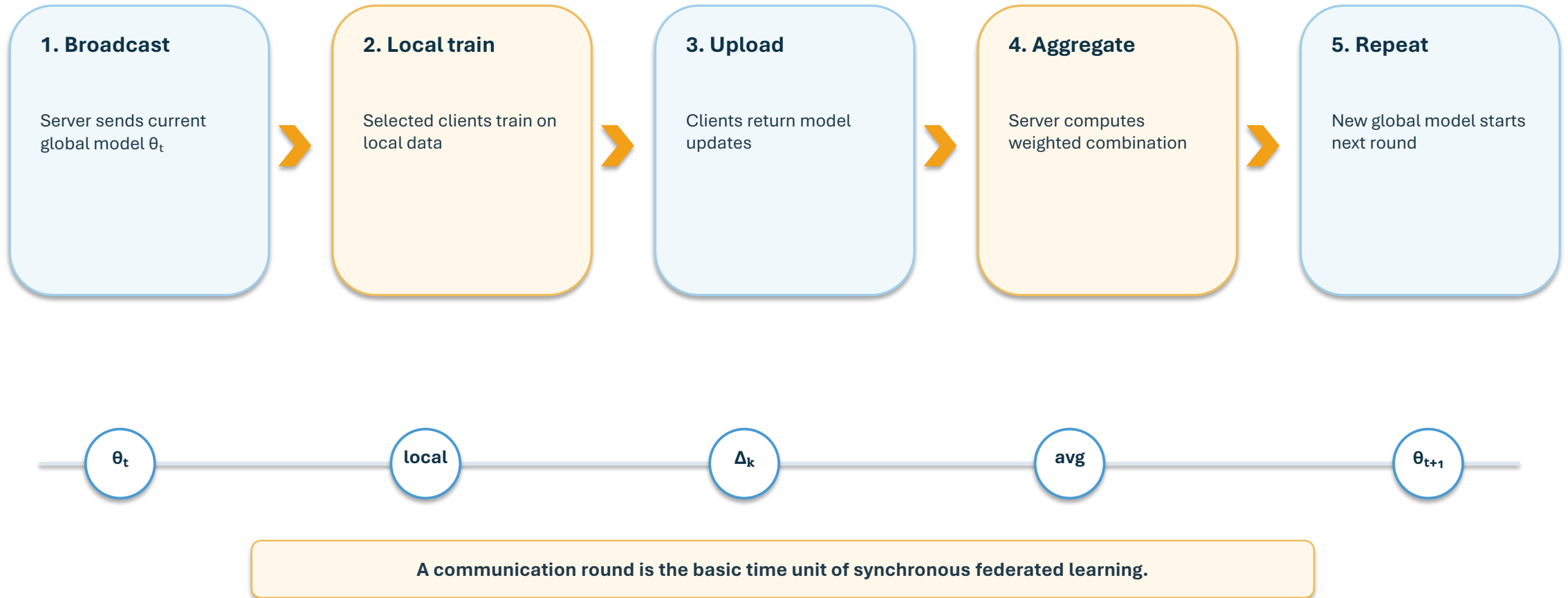
Reason about communication, client drift, and convergence behavior.

Today's path

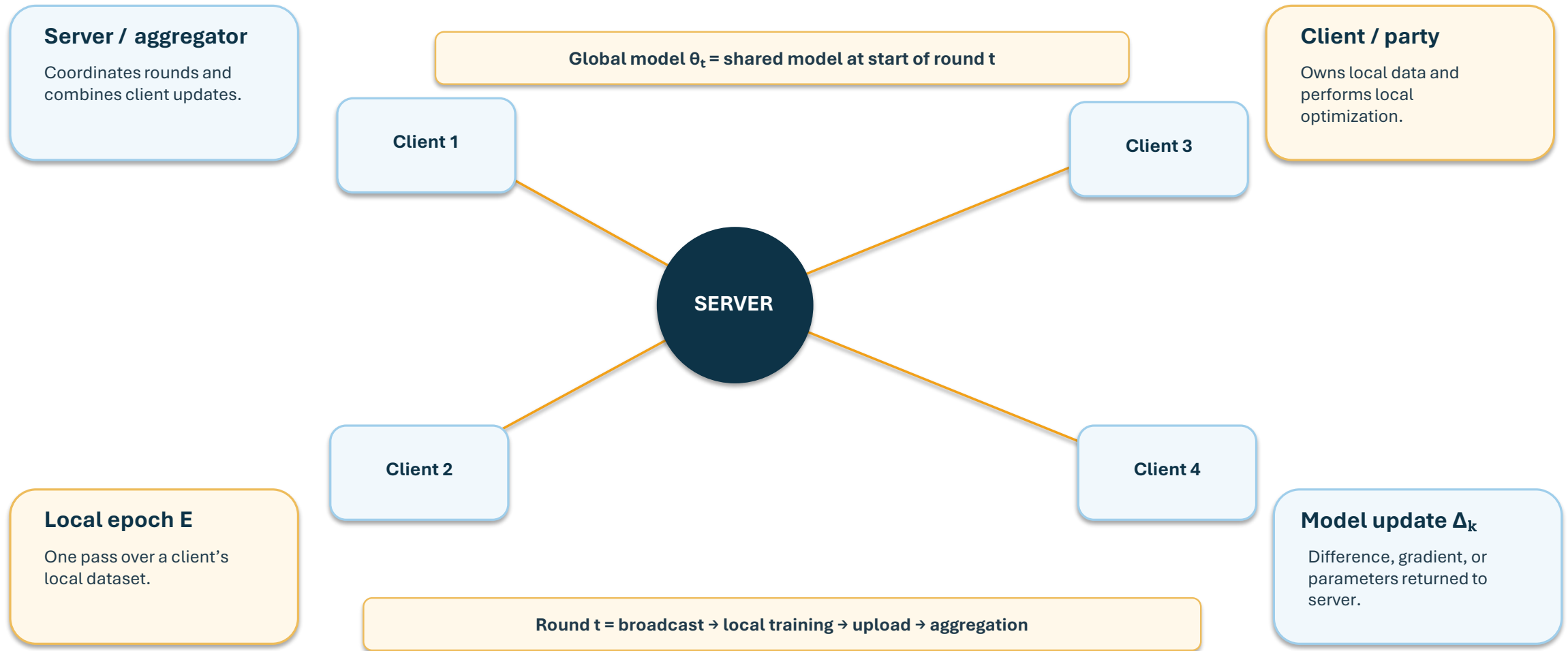


Goal: understand the baseline algorithm before studying heterogeneity-aware methods next week.

One Federated Learning Round



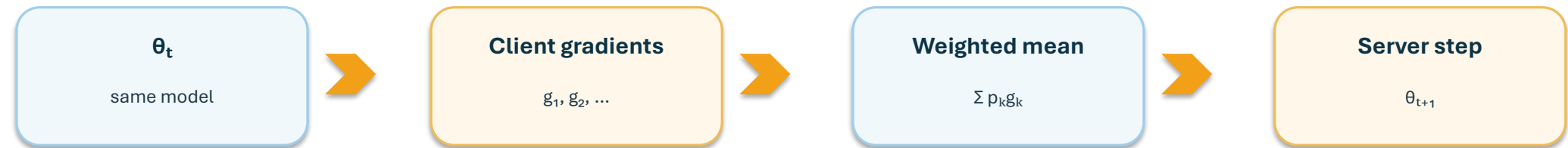
Core Architecture and Terminology



FedSGD: Aggregate Gradients

$$\theta_{t+1} = \theta_t - \eta \sum_k p_k g_k(\theta_t)$$

Each selected client computes a gradient at the same global model; the server averages the gradients and updates once.



Frequent synchronization

Clients communicate after each gradient computation.

Simple baseline

Closely resembles distributed mini-batch SGD.

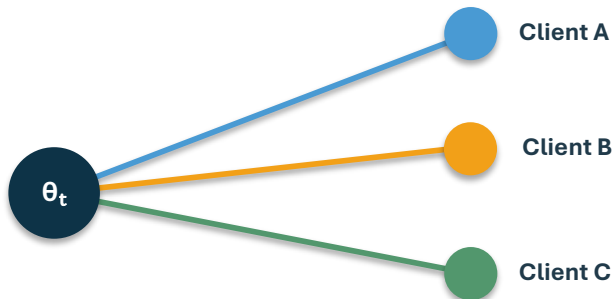
Main cost

Communication can dominate training time.

FedAvg: Local Updates Before Aggregation



What changes relative to FedSGD?



Local computation replaces communication

Instead of communicating every gradient step, each client performs multiple local steps before one upload.

Benefit and risk

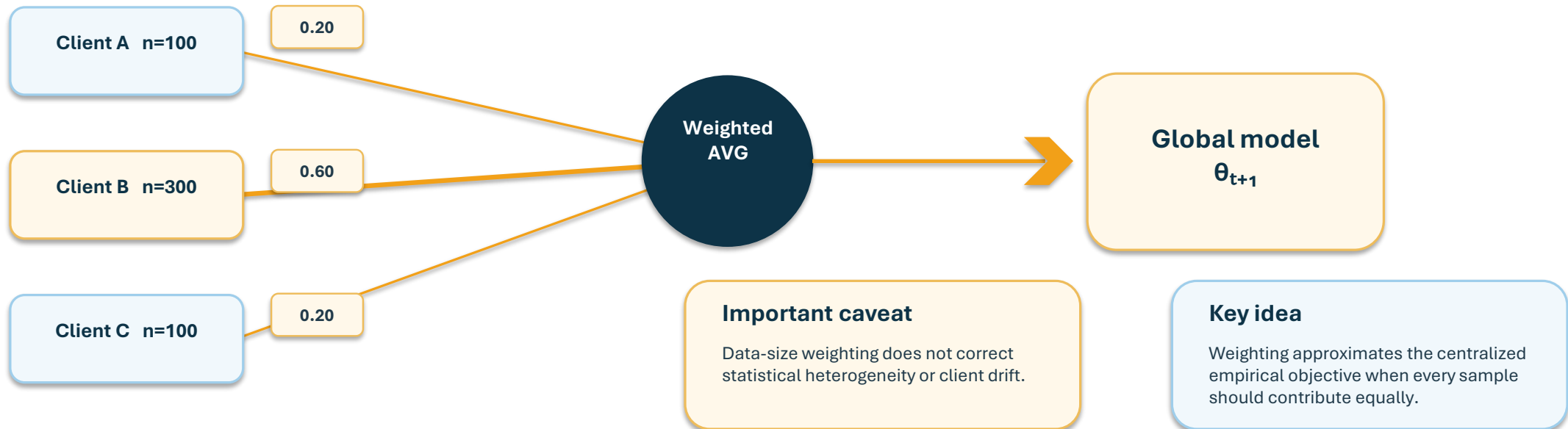
Communication drops substantially, but local models can drift apart when data are heterogeneous or E is large.

FedAvg = local mini-batch SGD + periodic weighted model averaging.

FedAvg Weighted Aggregation

$$\theta_{t+1} = \sum_{k \in S_t} (n_k / \sum_{j \in S_t} n_j) \cdot \theta_{t+1}^k$$

Clients with more local training samples receive larger weight under the standard data-size weighting rule.



Numerical Example

Suppose three selected clients return scalar model values after local training:

Client A

$$n_1 = 100$$

$$\theta^1 = 2.0$$

$$\text{weight} = 0.2$$

Client B

$$n_2 = 300$$

$$\theta^2 = 4.0$$

$$\text{weight} = 0.6$$

Client C

$$n_3 = 100$$

$$\theta^3 = 5.0$$

$$\text{weight} = 0.2$$

$$\theta_{t+1} = 0.2(2.0) + 0.6(4.0) + 0.2(5.0) = 3.8$$

Why 0.6 for Client B? Because it owns $300 / 500 = 60\%$ of the selected data.

Check: the weights sum to 1.

FedSGD vs. FedAvg

Property	FedSGD	FedAvg
Local work per round	One / few gradient computations	Multiple local SGD steps
Communication frequency	High	Lower
Client drift	Low	Can increase with local steps
Parallel role	Centralized-like gradient averaging	Local training + periodic averaging
Typical use	Conceptual baseline	Canonical FL baseline

FedAvg trades more local computation for fewer communication rounds.

Hyperparameters That Matter

Client fraction C

How many clients participate per round?

Local epochs E

How many local passes before upload?

Batch size B

How many local examples per SGD step?

Learning rate η

How aggressively is each local model updated?

Rounds T

How many communication cycles are run?

Aggregation rule

How are client updates combined or filtered?

These parameters interact:

$\uparrow E \rightarrow \downarrow \text{communication}$

$\uparrow E \rightarrow \uparrow \text{client drift risk}$

$\downarrow C \rightarrow \downarrow \text{round cost}$

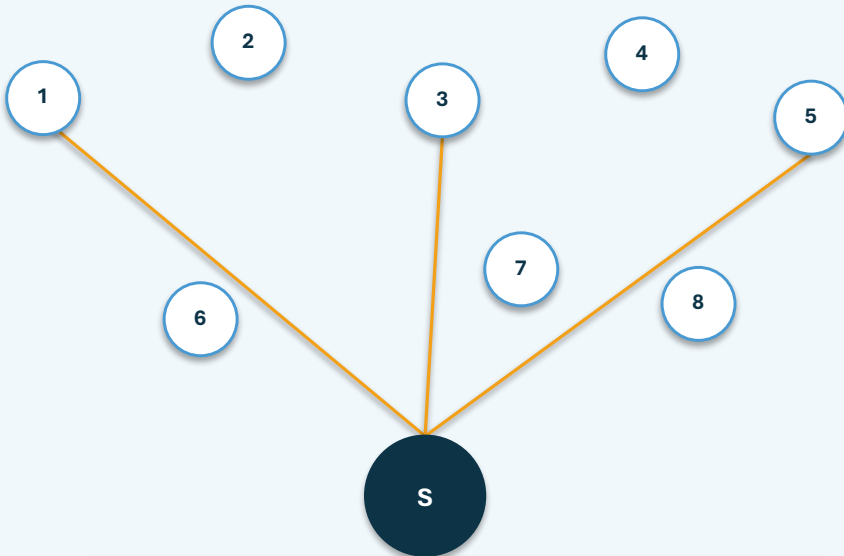
$\downarrow C \rightarrow \uparrow \text{sampling noise}$

There is no single 'best' setting: the correct choice depends on data heterogeneity, device resources, and communication cost.

Cross-Device vs. Cross-Silo FedAvg

Cross-device

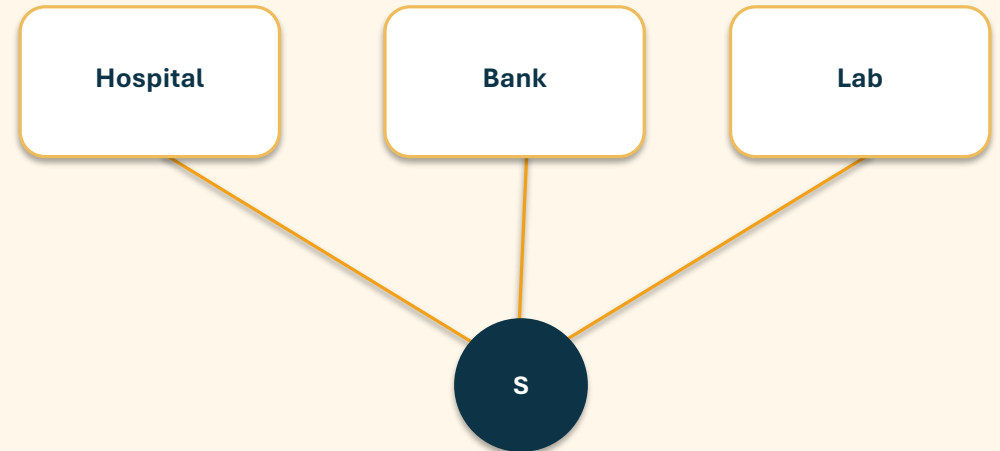
Many intermittently available clients; only a small subset usually participates in each round.



Examples: phones, wearables, IoT

Cross-silo

Small number of organizations or data centers; clients are more stable and often participate repeatedly.



Examples: hospitals, banks, research institutions

Cross-device emphasizes sampling and availability; cross-silo emphasizes organization-level heterogeneity and reliability.

Why FedAvg Works—and Why It Can Fail

Why it often works

Efficient communication

Many local steps per upload.

Simple baseline

Easy to implement and scale.

Strong empirical behavior

Works well when local objectives are reasonably aligned.

Where it breaks

Client drift

Local models move toward different optima.

Partial participation

Each round sees only part of the population.

Heterogeneity

Different data, systems, and update quality complicate averaging.

FedAvg is the canonical baseline—not the final answer. Weeks 4–6 study methods designed to fix its limitations.

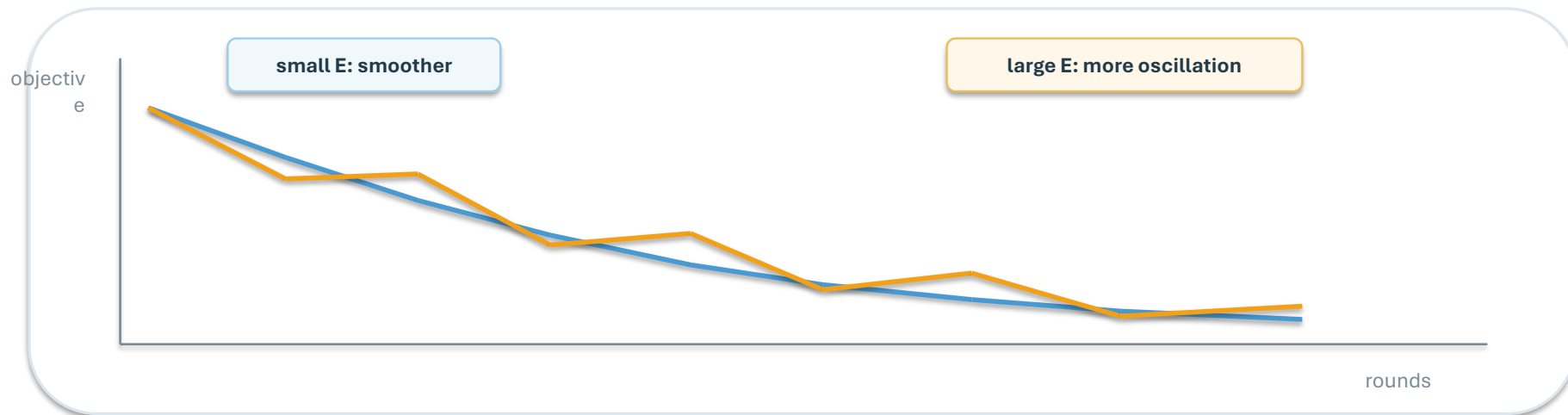
Convergence Intuition for FedAvg

Small E / frequent communication

Local models stay closer to the shared trajectory, but communication cost is high.

Large E / infrequent communication

Fewer rounds are needed, but local models can move farther apart under non-IID data.



Convergence is a balance among local steps, learning rate, heterogeneity, client sampling, and communication frequency.

In-Class Walkthrough: Trace One Round

Given

Three clients start from the same scalar parameter $\theta_t = 0$.

Local result

After local training: $\theta^A = 1.6$, $\theta^B = 2.4$, $\theta^C = 4.8$.

Data sizes

$n^A = 50$, $n^B = 150$, $n^C = 300$ samples.

Task

Compute the weighted FedAvg model and compare with a simple unweighted average.

Weights: 0.10, 0.30, 0.60

FedAvg = $0.1(1.6) + 0.3(2.4) + 0.6(4.8) = 3.76$

Unweighted mean = 2.93

Discussion: why does the weighted average move much closer to Client C?

Key Takeaways

- 1 **FedSGD aggregates gradients; FedAvg aggregates locally trained model updates.**
- 2 A communication round contains broadcast, local training, upload, and aggregation.
- 3 **Standard FedAvg weights each selected client by its local data size.**
- 4 More local work can reduce communication, but may increase client drift.
- 5 Client sampling, learning rate, batch size, local epochs, and heterogeneity jointly shape convergence.

Mental model: local SGD + periodic weighted averaging + partial client participation.

Reading & Preparation

Main textbook

Ludwig & Baracaldo (2022)
Review federated optimization and baseline training workflows.

Foundational paper

McMahan et al. (AISTATS 2017)
Communication-Efficient Learning of Deep Networks from Decentralized Data.

Prepare for Week 4 — Data Heterogeneity and Advanced Federated Optimization

- 1 Review IID vs. non-IID client data.
- 2 Think about why local models drift when client distributions differ.
- 3 Read the FedProx idea and identify what the proximal term changes.

Next week: why FedAvg degrades under heterogeneous local objectives—and how FedProx/SCAFFOLD-style ideas respond.

Q & A

Thank You.

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